

PHYSICS

1) Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$. Which of the following statements is **INCORRECT** ?

(1) $\vec{A} \cdot \vec{B} = 50$

(2) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$

(3) $|\vec{A}| = 5$

(4) $\vec{A} \cdot \vec{B} = 48$

2) Find the dot product of two vectors : $\vec{a} = 2\hat{i} + 3\hat{j} - 2\hat{k}$ & $\vec{b} = 5\hat{i} - 2\hat{j}$

(1) 4

(2) 16

(3) 10

(4) -4

3) Which of the following forces cannot be a resultant of 15 N force and 11 N force ?

(1) 2 N

(2) 10 N

(3) 14 N

(4) 5 N

4) Given that $0.2\hat{i} + 0.6\hat{j} + a\hat{k}$ is a unit vector. What is the value of a ?

(1) $\sqrt{0.3}$

(2) $\sqrt{0.4}$

(3) $\sqrt{0.6}$

(4) $\sqrt{0.8}$

5) Given $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ & $\vec{B} = \alpha\hat{i} + 3\hat{j} - 4\hat{k}$, the value of α so that $(2\vec{A} + 3\vec{B})$ is perpendicular to the $\vec{A}$ is :

(1) $\frac{11}{6}$

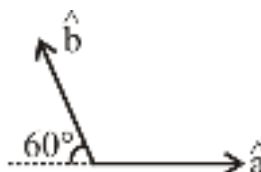
(2) $-\frac{11}{6}$

(3) $\frac{13}{2}$

(4) $-\frac{13}{2}$

6) If $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B}$ then angle between $\vec{A}$ and $\vec{B}$ should be :

- (1) 45°
- (2) 0°
- (3) 90°
- (4) Not possible for any θ



7) Magnitude of the resultant of two vectors shown is :-

- (1) $\sqrt{3}$
- (2) 1
- (3) 2
- (4) 0

8) Volume is :-

- (1) Scalar
- (2) Vector
- (3) Neither scalar nor vector
- (4) Both scalar and vector

9) If $\vec{A} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 5\hat{i} + y\hat{j} + z\hat{k}$ are parallel to each other then value of z and y will be :

- (1) 10, 5
- (2) 5, 10
- (3) 5, 5
- (4) 10, 10

10) The vector $\vec{P}$ makes 120° with the x-axis and the vector $\vec{Q}$ makes 30° with the y-axis. What is their resultant ?

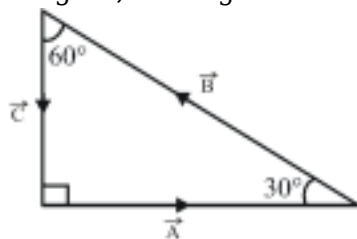
- (1) $P + Q$
- (2) $P - Q$
- (3) $\sqrt{P^2 + Q^2}$
- (4) $\sqrt{P^2 - Q^2}$

11) The magnitude of vector $2\hat{i} + 4\hat{j} + 5\hat{k}$ is :-

- (1) $4\sqrt{5}$

- (2) $3\sqrt{5}$
- (3) $5\sqrt{4}$
- (4) $5\sqrt{3}$

12) In the triangle shown in figure, the angle between vectors $\vec{A}$ and $\vec{B}$ is :-



- (1) 60°
- (2) 120°
- (3) 150°
- (4) 45°

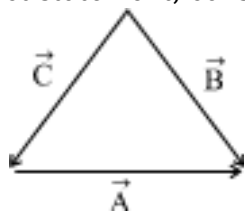
13) Which of following statements is/are correct :-

- (1) angle between parallel vectors is zero
- (2) angle between antiparallel vectors is 180°
- (3) parallel vectors have same unit vector and antiparallel vectors have opposite unit vector
- (4) All

14) Two forces, each of magnitude F have a resultant of the same magnitude F . The angle between the two forces is :-

- (1) 45°
- (2) 120°
- (3) 150°
- (4) 60°

15) Correct statement, considering figure is :-



- (1) $\vec{A} + \vec{B} = \vec{C}$
- (2) $\vec{B} + \vec{C} = \vec{A}$
- (3) $\vec{C} + \vec{A} = \vec{B}$
- (4) $\vec{A} + \vec{B} + \vec{C} = 0$

16) At what angle the two vectors of magnitude $(A + B)$ and $A - B$ must act, so that the resultant is

$$\sqrt{A^2 + B^2} ?$$

(1) $\cos^{-1} \frac{A^2 - B^2}{A^2 + B^2}$

(2) $\cos^{-1} \frac{A^2 + B^2}{B^2 - A^2}$

(3) $\cos^{-1} \frac{A^2 - B^2}{2(A^2 + B^2)}$

(4) $\cos^{-1} \frac{A^2 + B^2}{2(B^2 - A^2)}$

17) If $\frac{|\hat{a} + \hat{b}|}{|\hat{a} - \hat{b}|} = 1$, then angle between $\hat{a}$ and $\hat{b}$ is :-

(1) 0°

(2) 45°

(3) 90°

(4) 60°

18) If a vector $\vec{p}$ make angle α , β and γ respectively with the x, y and z axis. Then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$

(1) 0

(2) 1

(3) 2

(4) 3

19) If $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ the direction of cosines of the vector $\vec{A}$ are :-

(1) $\frac{2}{\sqrt{45}}, \frac{4}{\sqrt{45}}$ and $\frac{-5}{\sqrt{45}}$

(2) $\frac{1}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{3}{\sqrt{45}}$

(3) $\frac{4}{\sqrt{45}}, 0$ and $\frac{4}{\sqrt{45}}$

(4) $\frac{3}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{5}{\sqrt{45}}$

20) If $|\vec{a}| = 2$ unit, $|\vec{b}| = 5$ unit and $|\vec{a} \times \vec{b}| = 8$ unit, then, $\vec{a} \cdot \vec{b}$ is :-

(1) Zero

- (2) 2 unit
- (3) 4 unit
- (4) 6 unit

21) If $\vec{A}$, $\vec{B}$ & $\vec{C}$ are coplanar then which of the following will always be zero :-

- (1) $\vec{A} + \vec{B} - \vec{C}$
- (2) $(\vec{A} \times \vec{B}) \cdot \vec{C}$
- (3) $\vec{A} + \vec{B} + \vec{C}$
- (4) All of the above

22) Given that $A = B$. What is the angle between $(\vec{A} + \vec{B})$ and $(\vec{A} - \vec{B})$?

- (1) 30°
- (2) 60°
- (3) 90°
- (4) 180°

23) If the angle between the vectors $\vec{A}$ and $\vec{B}$ is θ , the value of the product $(\vec{B} \times \vec{A}) \cdot \vec{A}$ is equal to:-

- (1) $BA^2 \cos \theta$
- (2) $BA^2 \sin \theta$
- (3) $BA^2 \sin \theta \cos \theta$
- (4) zero

24) If adjacent sides $\vec{AB}$ and $\vec{AD}$ of parallelogram ABCD are $(2\hat{i} + 4\hat{j} - 5\hat{k})$ and $(\hat{i} + 2\hat{j} + 3\hat{k})$ respectively, then length of diagonal $\vec{AC}$ is :-

- (1) 3 unit
- (2) 4 unit
- (3) 5 unit
- (4) 7 unit

25) Two vectors $\vec{A}$ and $\vec{B}$ are inclined to each other at an angle θ . Which of the following is the unit vector perpendicular to both $\vec{A}$ and $\vec{B}$:-

- (1) $\frac{\vec{A} \times \vec{B}}{\vec{A} \cdot \vec{B}}$
- (2) $\frac{\hat{A} \times \hat{B}}{\sin \theta}$
- (3) $\frac{\vec{A} \times \vec{B}}{AB \sin \theta}$
- (4) $\frac{\hat{A} \times \vec{B}}{AB \sin \theta}$

26) A vector $\vec{A}$ is directed along 30° west of north direction and another vector $\vec{B}$ along 15° south of east. Their resultant cannot be in _____ direction.

- (1) North
- (2) East
- (3) North-East
- (4) South

27) A body is at rest under the action of three forces, two of which are $\vec{F}_1 = 4\hat{i}$, $\vec{F}_2 = 6\hat{j}$, then third force is :-

- (1) $4\hat{i} + 6\hat{j}$
- (2) $-4\hat{i} - 6\hat{j}$
- (3) $-4\hat{i} + 6\hat{j}$
- (4) $4\hat{i} - 6\hat{j}$

28) The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is rotated through an angle θ and doubled in magnitude, then it becomes $4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$. The positive value of x is :-

- (1) $\frac{2}{3}$
- (2) $\frac{1}{3}$
- (3) $\frac{2}{3}$
- (4) 2

29) If three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 12, 5 and 13 in magnitude such that $\vec{C} = \vec{A} + \vec{B}$ then angle between $\vec{A}$ & $\vec{B}$ is

- (1) 60°
- (2) 90°
- (3) 120°
- (4) 45°

30) If $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$ then the value of $|\vec{A} + \vec{B}|$ is : -

- (1) $(A^2 + B^2 + AB)^{1/2}$
- (2) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
- (3) $A + B$
- (4) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$

31) If vectors $5\hat{k}$ and $-a\hat{k}$ are equal vectors, then the value of a is : -

- (1) 5
- (2) 2
- (3) -3
- (4) -5

32) For a body, angular velocity $\vec{\omega} = \hat{i} - 2\hat{j} + 3\hat{k}$ and radius vector $\vec{r} = \hat{i} + \hat{j} + \hat{k}$, then its velocity ($\vec{v} = \vec{\omega} \times \vec{r}$) is :

- (1) $-5\hat{i} + 2\hat{j} + 3\hat{k}$
- (2) $-5\hat{i} + 2\hat{j} - 3\hat{k}$
- (3) $-5\hat{i} - 2\hat{j} + 3\hat{k}$
- (4) $-5\hat{i} - 2\hat{j} - 3\hat{k}$

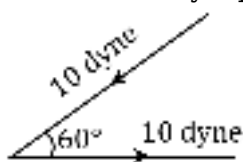
33) What is the component of $(3\hat{i} + 4\hat{j})$ along $(\hat{i} + \hat{j})$?

- (1) $\frac{1}{2}(\hat{j} + \hat{i})$
- (2) $\frac{3}{2}(\hat{j} + \hat{i})$
- (3) $\frac{5}{2}(\hat{j} + \hat{i})$
- (4) $\frac{7}{2}(\hat{j} + \hat{i})$

34) What is the maximum number of rectangular components into which a vector can be split in its own plane ?

- (1) 2
- (2) 3
- (3) 4
- (4) Infinite

35) Two forces each numerically equal to 10 dynes are acting as shown in the adjoining figure. Then

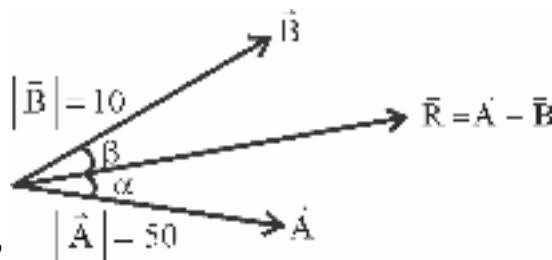


the resultant is :

- (1) 10 dyne
- (2) 20 dyne
- (3) $10\sqrt{3}$ dyne
- (4) 5 dyne

36) $\vec{A} = 4\hat{i} + 3\hat{j}$ and $\vec{B} = 4\hat{i} + 2\hat{j}$. Find a vector parallel to $\vec{A}$ but has magnitude five times that of $\vec{B}$.

- (1) $\sqrt{20}(2\hat{i} + 3\hat{j})$
- (2) $\sqrt{20}(4\hat{i} + 3\hat{j})$
- (3) $\sqrt{20}(2\hat{i} + \hat{j})$
- (4) $\sqrt{10}(2\hat{i} + \hat{j})$



37) Which of the following statement is true ?

- (1) $\beta > \alpha$
- (2) $\alpha > \beta$
- (3) $\alpha = \beta$
- (4) None

38) $\vec{A} = 2\hat{i} + \hat{j}$, $\vec{B} = 3\hat{j} - \hat{k}$ and $\vec{C} = 6\hat{i} - 2\hat{k}$ value of $\vec{A} - 2\vec{B} + 3\vec{C}$ would be -

- (1) $20\hat{i} + 5\hat{j} + 4\hat{k}$
- (2) $20\hat{i} - 5\hat{j} - 4\hat{k}$
- (3) $4\hat{i} + 5\hat{j} + 20\hat{k}$
- (4) $5\hat{i} + 4\hat{j} + 10\hat{k}$

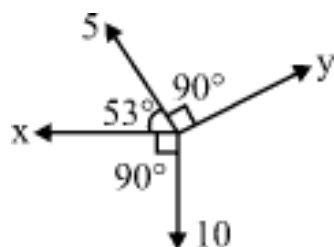
39) Find value of $|4\hat{A} + 3\hat{B}|$, if angle between $\hat{A}$ and $\hat{B}$ is 120°

- (1) $\sqrt{19}$
- (2) $\sqrt{37}$
- (3) $\sqrt{13}$
- (4) $\sqrt{25}$

40) The area of the parallelogram whose sides are represented by the vectors $\hat{j} + 3\hat{k}$ and $\hat{i} + 2\hat{j} - \hat{k}$ is :-

- (1) $\sqrt{61}$ sq. unit
- (2) $\sqrt{59}$ sq. unit
- (3) $\sqrt{49}$ sq. unit
- (4) $\sqrt{52}$ sq. unit

41) Find the magnitude of the unknown forces X and Y if sum of all forces is zero.



- (1) 4, 8
- (2) 5, 10
- (3) 4, 10
- (4) 5, 5

42) The X and Y components of vector $\vec{A}$ have numerical values 6 and 6 respectively and that of $(\vec{A} + \vec{B})$ have numerical values 10 and 9. What is the magnitude of $\vec{B}$:-

- (1) 2
- (2) 3
- (3) 4
- (4) 5

43) $|\vec{A} \times \vec{B}|^2 + |\vec{A} \cdot \vec{B}|^2$ is equal to :-

- (1) Zero
- (2) $A^2 B^2$
- (3) AB
- (4) $\sqrt{AB}$

44) If $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = 4\hat{i} + 4\hat{j} + 4\hat{k}$ and $\vec{R} = \vec{A} + \vec{B}$ then angle that $\vec{R}$ makes with $\vec{A}$ is :-

- (1) 37°
- (2) 0°
- (3) 180°
- (4) 45°

45) Given that $\vec{A} + \vec{B} + \vec{C} = \vec{0}$. Out of these three vectors two are equal in magnitude and the magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. Then the angles between vectors are :

- (1) $30^\circ, 60^\circ, 90^\circ$
- (2) $45^\circ, 45^\circ, 90^\circ$
- (3) $45^\circ, 60^\circ, 90^\circ$
- (4) $90^\circ, 135^\circ, 135^\circ$

1)

200 gm CaCO_3 heated and it decomposed into CaO and CO_2 . What is the weight of residue -(Atomic weight of $\text{Ca} = 40$)

- (1) 56 gm
- (2) 200 gm
- (3) 100 gm
- (4) 112 gm

2) Which of the following statement is correct :

- (1) 1 mole of electrons has 1.6×10^{-19} C of charge
- (2) 1 mole of electrons weighs 0.54 mg
- (3) 1 mole of electrons weighs 5.4 mg
- (4) 1 mole of electrons weighs 0.54 kg

3) Density of ozone relative to oxygen is under the same temperature and pressure

- (1) 1
- (2) 2
- (3) 1.5
- (4) 2.5

4) Two atoms are said to be isobars if :-

- (1) They have same atomic number but different mass number
- (2) They have same number of electrons but different number of neutrons.
- (3) They have same number of neutrons but different number of electrons.
- (4) Sum of the number of protons and neutrons is same but the number of protons is different.

5) Which contains least number of molecules:-

- (1) 1 g O_2
- (2) 1g H_2
- (3) 1 g O_3
- (4) 1g NH_3

6)

The law of conservation of mass holds good for all of the following except.

- (1) All chemical reactions
- (2) Endothermic reactions
- (3) Exothermic reactions
- (4) Nuclear reactions

7)

Select the **incorrect** statement

- (1) Density of O_2 gas at NTP is 1.429 gL^{-1} .
- (2) Number of molecules in 18 mL of water will be equal to avogadro number ($d = 1 \text{ g/mL}$)
- (3) The weight of 2g-atom of Mg($A_t = 24$) will be 48g.
- (4) Number of H atoms in 18 g water will be equal to avogadro's number.

8) A compound possesses 8% sulphur by mass. The least molecular mass of compound is :-

- (1) 200
- (2) 400
- (3) 155
- (4) 355

9)

A gaseous mixture contains oxygen and nitrogen in the ratio 1:4 by weight. Therefore, the ratio of the number of molecules is :-

- (1) 1:4
- (2) 1:8
- (3) 7:32
- (4) 3:16

10) The volume occupied by 0.1 mol of CO_2 at NTP:-

- (1) 22.4 L
- (2) 2.24 L
- (3) 0.224 L
- (4) 0.1 L

11) At NTP, 11.2 L of a gas weight 16 g. The vapour density of the gas is :-

- (1) 32
- (2) 40
- (3) 16
- (4) 8

12) If water sample are taken from sea, rivers or lake, they will be found to contain hydrogen and oxygen in the approximate ratio of 1 : 8. This indicates the law of

- (1) multiple proportion
- (2) definite proportion
- (3) reciprocal proportion
- (4) none of these

13) If mass of helium sample is 16 amu. How many atoms present in it?

- (1) 4
- (2) $4 \times N_A$
- (3) N_A
- (4) 2

14) The number of electrons in one molecule of CO_2 are :-

- (1) 22
- (2) 44
- (3) 66
- (4) 88

15) A mixture of 3 L of C_3H_8 and C_4H_{10} on complete combustion at $25^\circ C$ produce 10 L of CO_2 . What is the composition of gaseous mixture?

- (1) 2 L of C_3H_8 , 1 L of C_4H_{10}
- (2) 2 L of C_4H_{10} , 1 L of C_3H_8
- (3) 1.5 L of C_4H_{10} , 1.5 L of C_3H_8
- (4) 1.8 of C_3H_8 , 1.2 L of C_4H_{10}

16) Calculate the mass of lime (CaO) obtained by heating 200 kg of 90% pure lime stone ($CaCO_3$) :-

- (1) 104.4 kg
- (2) 105.4 kg
- (3) 212.8 kg
- (4) 100.8 kg

17)

Select the **correct** statement :

- (1) The weight of 1 molecule of water will be 18 g.
- (2) The number of C atoms in 1 mole $C_6H_{12}O_6$ will be $6N_A$.
- (3) 2 moles of H_2 at NTP occupy 4L volume.
- (4) Vapour density = 2 $\times$ mol. wt.

18) The total number of AlF_3 units in a sample of AlF_3 containing 3.01×10^{23} ions of F^- is:-

- (1) 9.0×10^{24}
- (2) 3.0×10^{24}
- (3) 7.5×10^{23}
- (4) 10^{23}

19) Equal volume of different molecular gases at any definite temperature and pressure have

- (1) Equal density
- (2) Equal mass
- (3) Equal no. of atoms
- (4) Equal no. of molecules

20) Ribose has the formula $C_5H_{10}O_5$. What is the mass of ribose sample containing 1.25×10^{24} carbon atoms :-

- (1) 23.9 g
- (2) 1.99 g
- (3) 150.1 g
- (4) 62.5 g

21) Which of the following is correct for a 4d-electron

- (1) $n = 4, \ell = 2, s = +\frac{1}{2}$
- (2) $n = 4, \ell = 1, s = -\frac{1}{2}$
- (3) $n = 4, \ell = 0, s = +\frac{1}{2}$
- (4) $n = 4, \ell = 3, s = +\frac{1}{2}$

22) Match **List-I** with **List-II** :

	List-I (quantum number)		List-II (Orbital)
(a)	$n = 2, \ell = 1$	(i)	2 s
(b)	$n = 3, \ell = 2$	(ii)	3 s
(c)	$n = 3, \ell = 0$	(iii)	2 p
(d)	$n = 2, \ell = 0$	(iv)	3 d

Choose the correct answer from the options given below :

- (1) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
- (2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (3) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
- (4) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

23) Quantum number which can not be calculated by schrodinger wave equation :-

- (1) n
- (2) ℓ
- (3) m
- (4) s

24) The orbital angular momentum and angular momentum (classical analogue) for the electron of

4s-orbital are respectively equal to :-

- (1) $\frac{\sqrt{12}h}{2\pi}$ and $\frac{h}{2\pi}$
- (2) zero and $\frac{2h}{\pi}$
- (3) $\sqrt{6}h$ and $\frac{2h}{\pi}$
- (4) $\sqrt{2}\frac{h}{2\pi}$ and $\frac{3h}{2\pi}$

25) Match the columns.

	Column-I		Column-II
(A)	Principal quantum number	(p)	Orientation of the orbital
(B)	Azimuthal quantum number	(q)	Energy and size of orbit
(C)	Magnetic quantum number	(r)	Spin of electron
(D)	Spin quantum number	(s)	Shape of the orbital

- (1) A - (q), B - (s), C - (p), D - (r)
- (2) A - (s), B - (q), C - (p), D - (r)
- (3) A - (q), B - (p), C - (s), D - (r)
- (4) A - (q), B - (s), C - (r), D - (p)

26) Which of the following is electronic configuration of Cu^{2+} ($Z = 29$)?

- (1) $[\text{Ar}]4s^1 3d^8$
- (2) $[\text{Ar}]4s^2 3d^{10} 4p^1$
- (3) $[\text{Ar}]4s^1 3d^{10}$
- (4) $[\text{Ar}]3d^9$

27) Which of the following metal ions will have maximum no. of unpaired electrons :-

- (1) Fe^{2+}
- (2) Co^{2+}
- (3) Ni^{2+}
- (4) Mn^{2+}

28) Ψ_{311} is :-

- (1) $3p_z$
- (2) $3d_{z^2}$
- (3) $3p_x$

(4) $3d_{xy}$

29) No. of unpaired electrons in Cu^{2+} is :-

- (1) Zero
- (2) 9
- (3) 3
- (4) 1

30) The orbital with zero orbital angular momentum is :-

- (1) s
- (2) p
- (3) d
- (4) f

31) A 3p-orbital has :-

- (1) Two radial nodes
- (2) Two non radial nodes
- (3) One radial and one angular node
- (4) One radial and two angular node

32) $3p_y$ orbital has one nodal plane, the nodal plane is :-

- (1) xy plane
- (2) xz plane
- (3) yz plane
- (4) All of above

33) In H atom if 'x' is the radius of the first bohr orbit, de-Broglie wavelength of an electron in 3^{rd} orbit is :-

- (1) $3\pi x$
- (2) $6\pi x$
- (3) $\frac{9x}{2}$
- (4) $\frac{x}{2}$

34) In chromium atom, find out the no. of unpaired electron with $n = 3$ and $m = +1$:-

- (1) 3
- (2) 2
- (3) 1
- (4) 0

35) Number of orbitals in 3rd orbit is :

- (1) 3
- (2) 9
- (3) 18
- (4) 36

36) Number of waves in 5th orbit :-

- (1) 4
- (2) 3
- (3) 5
- (4) 1

37) ${}_7\text{N}^{14} = 1s^2 2s^2 2p^3$, correct statement is :-

- (1) s-electrons = 2
- (2) p-electrons = 3
- (3) K-shell electrons = 4
- (4) L-shell electrons = 7

38) Which of the following transition neither shows absorption nor emission of energy in case of hydrogen atom :-

- (1) $3p_x \rightarrow 4p_x$
- (2) $4p_x \rightarrow 2p_y$
- (3) $3s \rightarrow 3d_{xy}$
- (4) All of above

39) "The exact path of electron in 3p-orbital cannot be determined. The above statement is based on :-

- (1) Hund's rule
- (2) Bohr's rule
- (3) de-Broglie concept
- (4) Uncertainty principle

40) The de-Broglie wavelength of a particle with mass 1 g and velocity 100 m sec^{-1} will be :-

- (1) $6.63 \times 10^{-33} \text{ m}$
- (2) $6.63 \times 10^{-34} \text{ m}$
- (3) $6.63 \times 10^{-35} \text{ m}$
- (4) $6.63 \times 10^{-30} \text{ m}$

41) What is the wavelength of a moving electron having kinetic energy (KE) is :-

$$(1) \lambda = \frac{h}{\sqrt{2mKE}}$$

$$(2) \lambda = \frac{h}{\sqrt{mKE}}$$

$$(3) \lambda = \sqrt{\frac{h}{2mKE}}$$

$$(4) \lambda = \frac{h}{2mKE}$$

42) How many electrons can fit in the orbital for which $n = 3$ and $l = 1$:-

- (1) 10
- (2) 14
- (3) 2
- (4) 6

43) For reaction $A + 2B \rightarrow C$. Find moles of C produced if 5 moles of A and 8 moles of B are initially taken :-

- (1) 5 mole
- (2) 2 mole
- (3) 4 mole
- (4) 8 mole

44) What quantity of lime stone on heating will give 44 kg of CO_2 gas :-

- (1) 1000 kg
- (2) 100 kg
- (3) 44 kg
- (4) 56 kg

45) Calculate the total volume of all gases at STP produced by complete decomposition of 34 g NH_3 , according to following reaction, $2NH_3 \rightarrow N_2 + 3H_2(g)$:-

- (1) 22.4 L
- (2) 44.8 L
- (3) 67.2 L
- (4) 89.6 L

BIOLOGY

1) Which of the following option is correct regarding animal kingdom?

S.No.	Phylum	Symmetry	Example	Characteristic property
1	Coelenterata	Bilateral	Hydra	Aquatic, Marine

2	Annelida	Bilateral	Ancylostoma	Hooks and Suckers present
3	Platyhelminthes	Bilateral	Planaria	High regeneration capacity
4	Mollusca	Radial	Pinctada	Aquatic

- (1) 1
(2) 2
(3) 3
(4) 4

2) Read the following statements (a to e) carefully :-

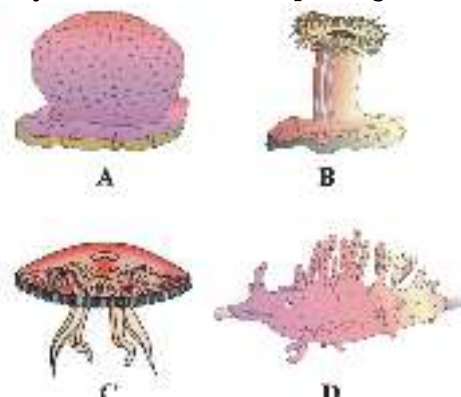
- (a) Triploplastic
(b) Bilateral symmetric
(c) Organ system organisation
(d) Pseudocoelomate
(e) Metameric segmentation

Which of the above statement(s) is/are correct for phylum Aschelminthes

- (1) a and b only
(2) a, b, c and e
(3) a, b, c and d
(4) a, b, c, d and e

3) Identify the following four animals

(A, B, C and D) given below. Which one of these is not correctly identified in the option given along



with its correct taxonomic group and their common name ?

Option	Fig.	Name of animal	Taxonomic group	Common Name
1	B	Adamsia	Ctenophora	Star coral
2	D	Spongilla	Porifera	Fresh water sponge
3	C	Aurelia	Coelenterata	Jelly fish
4	A	Euspongia	Porifera	Bath sponge

- (1) 1
(2) 2
(3) 3
(4) 4

4) Given below are two statements :

Statement-I : Body of *Aplysia* is segmented with a distinct head, muscular foot and visceral hump.

Statement-II : The space between the visceral hump and the mantle is called the mantle cavity.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement-I** and **Statement-II** are incorrect.
- (2) **Statement-I** is correct but **Statement-II** is incorrect.
- (3) **Statement-I** is incorrect but **Statement-II** is correct.
- (4) Both **Statement-I** and **Statement-II** are correct.

5) Read the following statements and select the correct option.

Statement-A: Sponges exhibit cellular level of organisation and some division of labour occur among the cells.

Statement-B: When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry

- (1) Both statements are correct
- (2) Both statements are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

6) **Assertion (A) :** In sponges, cells are loosely aggregated and show some division of labour, while in coelenterates, cells performing similar functions are arranged into tissues.

Reason (R) : Sponges exhibit a tissue level of organisation, whereas coelenterates exhibit a cellular level of organisation.

- (1) A is false but R is true.
- (2) Both A and R are true and R is the correct explanation of A.
- (3) Both A and R are true but R is NOT the correct explanation of A.
- (4) A is true but R is false.

7) **Assertion (A):** Coelenterates are diploblastic animals with two embryonic layers, ectoderm and endoderm, and an undifferentiated layer, mesoglea, in between.

Reason (R): Coelenterates exhibit bilateral symmetry as their body can be divided into identical left and right halves in only one plane.

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is not the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

8) Identify the correct match for locomotory structure of animals :

- (1) Insects - Paired jointed legs
- (2) Annelids - Muscular foot
- (3) Echinodermata - Parapodia
- (4) Cnidarians - Comb plates

9)

Among vertebrates which of the following do **not** exclusively respire through lungs ?

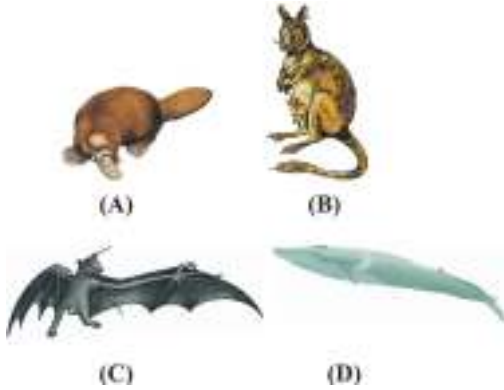
- (1) Birds
- (2) Mammals
- (3) Reptiles
- (4) Amphibians



10) The structure given below is present in which animal ?

- (1) *Scypha*
- (2) *Ctenoplana*
- (3) *Physalia*
- (4) *Taenia*

11) Identified the organisms A, B, C and D.



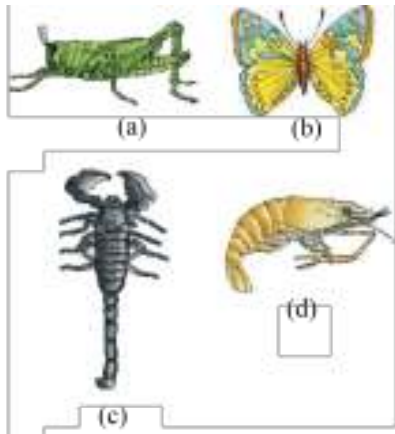
- (1) Balaenoptera, Macropus, Pteropus & Ornithorhynchus
- (2) Ornithorhynchus, Macropus, Pteropus & Balaenoptera
- (3) Ornithorhynchus, Macropus, Balaenoptera & Pteropus
- (4) Ornithorhynchus, Balaenoptera, Pteropus & Macropus

12) Radula, shell, mantle cavity present in :-

- (1) *Echinus*
- (2) *Limulus*
- (3) Sea hare
- (4) *Ctenoplana*

13) Identify the incorrect statement regarding Osteichthyes :-

- (1) Skin is covered with placoid scales
- (2) Air bladder is present
- (3) Four pairs of gills with operculum
- (4) Skin with cycloid/ctenoid scales



14) Identify the correct names of (a), (b), (c), & (d) in the figure :-

- (1) Locust, Prawn, Scorpion, Spider
- (2) Prawn, Butterfly, Scorpion, Locust
- (3) Locust, Butterfly, Scorpion, Prawn
- (4) Butterfly, Moth, Prawn, Aphid

15) The unique characteristic feature of sponges:

- (1) Presence of water vascular system
- (2) Presence of cnidoblast cells
- (3) Presence of choanocyte cells
- (4) Presence of diploblastic body

16) Select the correct option for *Balanoglossus* :-

- (1) Fresh water animal
- (2) True notochord is present
- (3) Excretion by proboscis gland
- (4) Triploblastic and circulatory system is close type

17) Which is not a sponges?

- (1) Sycon
- (2) Spongilla
- (3) Scypha
- (4) Sea anemone

18) In which of the following organisms notochord is persistent throughout the life?

- (1) *Herdmania*
- (2) *Branchiostoma*
- (3) *Ascidia*
- (4) *Balanoglossus*

19) Identify the correctly matched option:

- (1) Frog - Two chambered heart
- (2) Fish - Warm blooded animals
- (3) Toad - Three chambered heart
- (4) *Hyla* - Limbless amphibian

20) Bilateral symmetry and segmentation is found in:

- (1) Annelida, Arthropoda, Mollusca
- (2) Arthropoda, Mollusca, Echinodermata
- (3) Annelida, Arthropoda, Chordata
- (4) Arthropoda, Echinodermata, Chordata

21) The animals in which mesoderm is present as scattered pouches in between the ectoderm and endoderm are

- (1) Anopheles and Pila
- (2) Wuchereria and Ancylostoma
- (3) Ascaris and Taenia
- (4) Pleurobrachia and Ctenoplana

22) Which of the following correctly matched ?

- (1) Notochord - Mesodermal
- (2) Metamerism - Platyhelminthes
- (3) Triploblastic - Ctenophores
- (4) Pseudocoelom - Molluscs

23) Which one of the following groups of animals is correctly matched with its one characteristic feature without even a single exception ?

- (1) Reptilia : possess 3-chambered heart with one incompletely divided ventricle
- (2) Chordata : possess a mouth provided with an upper and a lower jaw
- (3) Chondrichthyes : possess cartilaginous endoskeleton
- (4) Mammalia : give birth to young ones

24) Which pair of characters are true for ctenophora?

- (1) Bioluminescence, Comb-plates
- (2) Comb-plates, direct development

(3) Exclusively marine, water-canal system

(4) Indirect development, dimorphic

25) Homeothermes, 4-chambered heart, body hairs, ear-pinna, are not true for

(1) *Pteropus*

(2) *Macaca*

(3) *Corvus*

(4) *Camelus*

26) Excretory organ of *Balanoglossus* is-

(1) Gills

(2) Solenocytes / Flame cells

(3) Proboscis-gland

(4) Collar

27) Gastro-vascular cavity in Cnidaria is also known as

(1) Coelom

(2) Coelenteron

(3) Pseudocoelom

(4) Schizocoelom

28) First phylum to have complete digestive tract can be represented by :-

(1) *Hydra*

(2) *Taenia*

(3) *Ancylostoma*

(4) *Pheretima*



29) Cross section of an animal body is given below. animals have above cross section ?

Which of the following

(1) *Fasciola*

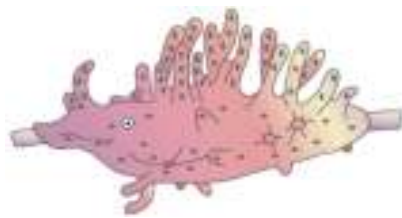
(2) *Pleurobrachia*

(3) *Ascaris*

(4) *Nereis*

30) Animals of phylum platyhelminthes are called flat worm because :-

- (1) Of circular cross section
- (2) These are laterally flat
- (3) These are antero-posteriorly flat
- (4) These have dorso-ventrally flattened body



31) Observe the given figure :

Select correct answer for it :

- (1) It is a fresh water sponge
- (2) Animal body is porous
- (3) It has radial symmetry
- (4) Both 1 & 2

32) The closed circulatory system first time developed in?

- (1) Phylum Annelida
- (2) Phylum Porifera
- (3) Phylum Mollusca
- (4) Phylum Arthropoda

33) Which of the following have only one opening to outside, serve as both mouth and Anus.

- (1) Aschelminthes
- (2) Sponges
- (3) Echinodermates
- (4) Platyhelminthes

34) Which of the following not correctly matched :-

- (1) Limulus - King crab
- (2) Locusta - Locust
- (3) Bombyx-Lac insect
- (4) Apis-Honey bee

35) Mouth part contains a file-like rasping organ for feeding called radulla-characteristics of :-

- (1) Arthropoda
- (2) Mollusca
- (3) Hemichordata
- (4) Echinodermata

36) The most distinctive feature of Echinodermata is :-

- (1) Fertilization is usually internal
- (2) Canal system
- (3) Development is direct without larval stage
- (4) Water vascular system

37) Chordates possess :-

- (1) Notochord absent
- (2) Post anal Tail absent
- (3) Central nervous system is dorsal, hollow and single
- (4) Heart present dorsally

38) Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata and Chordata are :-

- (1) Coelomates
- (2) Pseudocoelomates
- (3) Acoelomates
- (4) Coelomates and Acoelomates both

39) "Fresh water sponge" is :-

- (1) *Cliona*
- (2) *Spongilla*
- (3) *Euspongia*
- (4) *Hyalonema*

40) Endoskeleton in sponges are made up of :-

- (1) Cartilage
- (2) Spicules or spongin fibres
- (3) Bones
- (4) All of the above

41) Frog's eye are protected under water by the presence of :-

- (1) Retina
- (2) Upper eyelid
- (3) Nictitating membrane
- (4) Lower eyelid

42) Bidder's canal in frog is present in :-

- (1) Testes
- (2) Ovary
- (3) Kidney
- (4) Brain

43) Number of cranial nerves and spinal nerves in frog is :-

- (1) 10 and 20
- (2) 10 and 10
- (3) 20 and 10
- (4) 20 and 20

44) A mature female frog can lay ____.

- (1) 2500 to 3000 ova
- (2) 25 to 30 ova
- (3) 250 to 300 ova
- (4) 125 to 130 ova

45) The fore limb of male frog is distinct from female frog due to the presence of :-

- (1) Fore arm
- (2) Thumb
- (3) Web
- (4) Copulatory pad

46) Protoplasm is made up of-

- (1) Cytoplasm & nucleus
- (2) Cytoplasm & nucleolus
- (3) E.R. and Golgi body
- (4) Mitochondria & plastids

47) **Assertion :-** Content of nucleolus is continuous with the rest of the nucleoplasm.

Reason :- Nucleolus is not covered with any kind of membrane.

- (1) Assertion and Reason are true and Reason explains assertion correctly.
- (2) Assertion and Reason are true and Reason do not explain assertion correctly.
- (3) Assertion and Reason both are true.
- (4) Assertion is true but Reason is false.

48) Which of the following arrangements of microtubules is correct for flagellum ?

- (1) 9 peripheral singlets + 2 central doublets
- (2) 9 peripheral triplets + 2 central singlets
- (3) 9 peripheral doublets + 2 central singlets
- (4) 9 peripheral doublets + 2 central triplets

49) **Assertion (A):** Mitochondria are the sites of aerobic respiration.

Reason (R): They produce ATP by the process of photophosphorylation.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.
- (2) **(A)** is correct but **(R)** is not correct.
- (3) **(A)** is incorrect but **(R)** is correct.
- (4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**.

50) The accompanying diagram represents a chromosome Identify the structure A, B & type of



chromosome C :- C – Type of chromosome

- (1) A-Satellite, B-Primary constriction, C-Acrocentric
- (2) A-Satellite, B-Secondary constriction, C-Metacentric
- (3) A-Satellite, B-Centromere, C-Telocentric
- (4) A-Satellite, B-Centromere, C-Submetacentric

51) Flagella is responsible for :-

- (1) Cell movement
- (2) Movement of surrounding fluid
- (3) Maintenance of shape of cell
- (4) Formation of the basal body

52) Due to the absence of which of the following cell organelles, both prokaryotes and plant cells are differ from an animal cell ?

- (1) 80 s ribosomes
- (2) Mesosome
- (3) Centrioles
- (4) 70 s ribosomes

53) Movement of polar molecules against the concentration gradient across plasma membrane is possible through :-

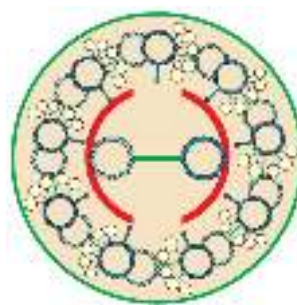
- (1) Diffusion
- (2) Osmosis
- (3) Carrier pump proteins
- (4) All of above

54) How many of the following are not included in endomembrane system?

Endoplasmic Reticulum, Golgi complex, Ribosomes, Lysosome, Mitochondria, Chloroplast, Vacuoles.

- (1) 2

- (2) 3
- (3) 4
- (4) 5



55) Which of the following is not correct about given diagram.

- (1) Arrangement of microtubules are 9 + 2.
- (2) Central microtubules are not include in axoneme.
- (3) Nine radial spoke are present.
- (4) Internal structure of cilia or flagella.

56) Chromosome, which has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm is called -

- (1) Submetacentric
- (2) Acrocentric
- (3) Metacentric
- (4) Telocentric

57) Chloroplast is different from mitochondria in having?

- (1) Cristae
- (2) Aerobic respiration
- (3) Single, circular DNA
- (4) Granum

58) An interconnected membranous network of the cell composed of vesicles, flattened sacs and tubules is

- (1) Mitochondria
- (2) Endoplasmic reticulum
- (3) Lysosomes
- (4) Nucleus

59) According to the fluid mosaic model, the quasi-fluid nature of lipid enables lateral movement of protein within the overall lipid bilayer. This ability to move within the membrane is measured as its

- (1) Mobility
- (2) Fluidity
- (3) Movability

(4) All of the above

60) Number of chloroplasts present in a *Chlamydomonas* and mesophyll cell respectively are :

- (1) 20 and 30 to 40 per cell
- (2) 1 and 20 to 40 per cell
- (3) 100 and 200 per cell
- (4) 20 and one to many per cell

61) Chromatin composed of :-

- (1) DNA and Histones Only
- (2) DNA and RNA Only
- (3) DNA, RNA and Histones Only
- (4) DNA, RNA, Histones and Non-histones

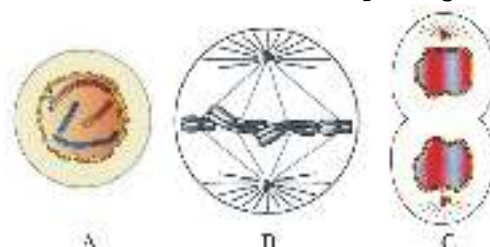
62) What is the correct arrangement of Lipid molecules in the cell membrane

- (1) Polar head → Outside, non - polar tails → Inner part
- (2) Non - polar head → Outside, Polar tail → Inner side
- (3) Polar tail → Outside, non - polar head → Inner side
- (4) Polar tail → inner side, non - polar head → outer side

63) If mitotically dividing diploid cell has initial amount of DNA as 5 Pg in G_1 stage, that is distributed among its 8 chromosomes then which of the following assumption is correct ?

- (1) G_2 - phase - 16 chromosomes with 10 Pg DNA.
- (2) Late prophase - 4 chromosomes with 10 Pg DNA.
- (3) Anaphasic cell - 16 chromosomes with 10 Pg DNA.
- (4) S stage-4 chromosomes with 2.5 Pg DNA.

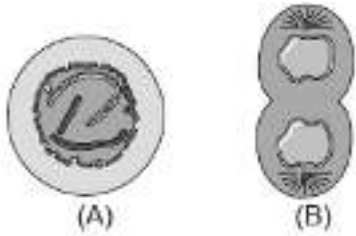
64) The figures below shows 3 phases of mitosis select the option giving correct identification



together with the correct event ?

- (1) C-Telophase-Nuclear envelope assembles around the chromosome clusters.
- (2) B-Anaphase-Segregation of homologous chromosomes.
- (3) A-Prophase-Chromosomes get fully condensed.
- (4) C-Metaphase-Condensation of chromatin to form chromosome

65) The given diagram (A) and (B) represents which phase of cell-division?



- (1) (A) - Prophase, (B) - Metaphase
- (2) (A) - Metaphase, (B) - Anaphase
- (3) (A) - Prophase, (B) - Telophase
- (4) (A) - Anaphase, (B) - Telophase

66) Crossing over is an enzyme-mediated process, The enzyme is :-

- (1) DNA ligase
- (2) Recombinase
- (3) RNAase
- (4) DNAase

67) The complete disintegration of nuclear envelope in a cell cycle marks the :-

- (1) Start of prophase of mitosis
- (2) Start of metaphase of mitosis
- (3) End of anaphase of mitosis
- (4) Start of telophase of mitosis

68) The process of crossing over occurs between :-

- (1) Sister chromatids of homologous chromosome
- (2) Non-sister chromatids of homologous chromosome
- (3) Sister chromatids of non homologous chromosome
- (4) Non-sister chromatids of non homologous chromosome

69) You have studied mitosis in onion root tip cells. It has 16 chromosomes in each cell. Can you tell how many chromosome will the cell have at G_1 -phase, metaphase, anaphase and just after M-phase respectively?

- (1) 16, 16, 32, 16
- (2) 16, 32, 32, 16
- (3) 16, 16, 32, 32
- (4) 16, 16, 16, 16

70) The complex formed by a pair of synapsed homologous chromosomes is called :-

- (1) Bivalent
- (2) Pentavalent
- (3) Tetrad

(4) Both 1 & 3

71) A-Condensation of chromatin materials

B-Nuclear envelope reform

C-Attachment of spindle fibres to kinetochore

D-Movement of sister chromatids toward poles

Choose the correct sequence of events given above (A to D) for mitosis :-

(1) A→B→C→D

(2) A→C→D→B

(3) A→C→B→D

(4) A→D→C→B

72) Choose the correct option for the stage at which morphology of chromosome is easily studied

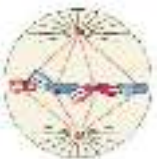
(1)



(2)



(3)



(4)



73) The correct sequence in cell cycle is :

(1) S - M - G₁ - G₂

(2) S - G₁ - G₂ - M

(3) G₁ - S - G₂ - M

(4) M - G₁ - G₂ - S

74) Liquid endosperm in coconut results due to :

(1) Karyokinesis is not followed by cytokinesis

(2) Karyokinesis followed by cytokinesis

(3) Cytokinesis followed by karyokinesis

(4) Cytokinesis not followed by karyokinesis

75) All the following are the significances of equational division except :-

- (1) Maintenance of the same number of chromosomes cell to cell
- (2) Growth of multicellular organisms and cell repair/ replacement
- (3) Conserve specific constant number of chromosomes generation to generation
- (4) Restoration of nucleo-cytoplasmic ratio of cells

76) The stage between two meiotic division is called interkinesis and :-

- (1) is long lived
- (2) involve DNA replication.
- (3) is short lived
- (4) Interval between prophase-I and metaphase-I

77) Match the column-I with column-II

Column-I		Column-II	
(A)	S phase	(i)	Most of organelles duplication
(B)	G ₁ phase	(ii)	Tubulin proteins are synthesised
(C)	Anaphase	(iii)	DNA Replication
(D)	G ₂ phase	(iv)	Centromere split

- (1) A-ii, B-iii, C-iv, D-i
- (2) A-i, B-iv, C-ii, D-iii
- (3) A-iii, B-i, C-iv, D-ii
- (4) A-iii, B-ii, C-i, D-iv

78) Given below are two statements : **Statements-I** : Membrane-bound organelles of the endomembrane system coordinate cellular functions.

Statement-II : Mitochondria and chloroplasts are not considered a part of the endomembrane system.

In the light of the above statements choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
- (2) Both **Statement-I** and **Statement-II** are correct
- (3) Both **Statement-I** and **Statement-II** are incorrect
- (4) **Statement-I** is correct but **Statement-II** is incorrect

79) Which one is correct about bivalent ?

- (i) Bivalents are tetrads
- (ii) A bivalent means 4 chromatids and 2 centromeres.
- (iii) One bivalent consists of 2 homologous chromosomes.

- (1) (i),(ii),and (iii)
- (2) (iii) only

- (3) only (ii) and (iii)
- (4) (i) only

80) **Assertion:** Meiosis is responsible for conservation of chromosome number of each species in sexually reproducing organisms.

Reason : Number of chromosomes reduced to half in gametes that are produced by meiosis.

- (1) Assertion and Reason are true and Reason explains assertion correctly
- (2) Assertion and Reason are true and Reason do not explain assertion correctly
- (3) Assertion and Reason both are true
- (4) Assertion is true but Reason is false

81) **Statement I :-** Chromosome cluster at opposite spindle poles and their identity is lost as discrete elements during metaphase.

Statement II :- Nucleolus, golgi complex and ER reform during telophase.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

82) Which of the following is correct regarding mitochondria?

- (1) Inner membrane more permeable than outer membrane.
- (2) New mitochondria is originated from pre-existing mitochondria by fission like division.
- (3) Mitochondria have single stranded circular DNA.
- (4) Matrix of mitochondria possess 80s ribosome.

83) Which one of the following statement is incorrect for nucleolus?

- (1) Related with NOR
- (2) Nucleolus is spherical structure present in the nucleoplasm.
- (3) Nucleolus is a site for synthesis of active Ribosomal protein
- (4) The content of nucleolus is continuous with the rest of the nucleoplasm.

84) **Assertion:** During zygotene stage chromosomes starts pairing together and this process is called synapsis.

Reason: Zygotene is marked by formation of complex structure called synaptonemal complex.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

85) How many cellular components in the list given below are non membrane bound cytoplasmic components of plant cell?

Nucleolus, Lysosome, Centrioles, Plastids, Ribosomes, Vacuole

- (1) Three
- (2) Four
- (3) Two
- (4) One

86) Identify the following cell organelle show below as well as its structure and function and select

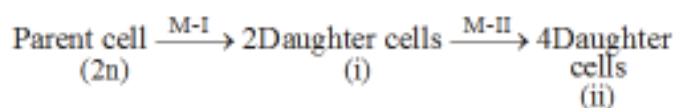


the right option for the two together :-

	Structure	Functions
(1)	Double membrane bound	Protein synthesis
(2)	Network of protein filaments	Mechanical support
(3)	Many flat, disc shaped - cisterne	Glycolipids & glycoproteins synthesis
(4)	Granular (Attached ribosomes)	Protein synthesis

- (1) 1
- (2) 2
- (3) 3
- (4) 4

87) What does(i) and (ii) represent in the given flowchart ? ($n \rightarrow$ number of chromosomes)



- (1) (i) = $2n$, (ii) = n
- (2) (i) = n , (ii) = n
- (3) (i) = n , (ii) = $2n$
- (4) (i) = $2n$, (ii) = $2n$

88) Section of cilia / flagella shows :-

	Peripheral Microtubules (Doublet)	Central microtubules (singlet)	Radial spoke	Central Sheath
(1)	$9 + 0$	2	8	1
(2)	$9 + 2$	$9 + 0$	9	1

(3)	9	2	9	1
(4)	3	6	9	1

(1) 1

(2) 2

(3) 3

(4) 4

89) During an experiment with onion root tips, A student stained and observed a slide under a microscope. In some cells, the nuclear membrane seemed to have disappeared and the chromatin fibres had condensed into thick thread-like structures aligned at the centre of the cell, which stage of cell division is likely being observed?

(1) Prophase

(2) Metaphase

(3) Telophase

(4) Anaphase

90) **Statement-A:** Mitochondria play a crucial role in cellular respiration as they are the primary sites where energy in the form of ATP is generated.

Statement-B: Mitochondria possess their own DNA and ribosomes, allowing them to synthesize some of their proteins independently.

(1) Both statements are correct.

(2) Both statements are not correct.

(3) Only statement-A is correct.

(4) Only statement-B is correct.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	1	3	1	4	2	1	2	1	2	3	4	2	3	4	3	3	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	4	4	2	4	2	4	2	1	4	1	4	1	1	2	1	2	3	2
Q.	41	42	43	44	45															
A.	2	4	2	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	3	4	3	4	4	2	3	2	3	2	1	1	1	4	2	4	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	4	2	1	4	4	3	4	1	3	2	2	3	2	3	2	3	4	1
Q.	86	87	88	89	90															
A.	1	3	3	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	1	3	1	4	3	1	4	3	2	3	1	3	3	3	4	2	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	1	3	3	2	3	3	4	4	1	4	3	2	4	3	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	1	4	1	1	3	2	2	1	3	3	2	2	1	4	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	3	1	3	2	2	2	1	4	2	3	3	1	3	3	3	2	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	3	1	4	3	2	3	2	1										

SOLUTIONS

PHYSICS

$$1) |\vec{A}| = \sqrt{3^2 + 4^2} = 5$$

$$|\vec{B}| = \sqrt{6^2 + 8^2} = 10$$

$$\vec{A} \cdot \vec{B} = (3\hat{i} + 4\hat{j}) \cdot (6\hat{i} + 8\hat{j})$$

$$= 18 + 32 = 50$$

$$\frac{|\vec{A}|}{|\vec{B}|} = \frac{5}{10} = \frac{1}{2}$$

$$2) \vec{a} \cdot \vec{b} (2\hat{i} + 3\hat{j} - 2\hat{k}) \cdot (5\hat{i} - 2\hat{j})$$

$$= 2 \times 5 + 3(-2) = 4$$

$$3) \vec{A} + \vec{B} = \vec{R}$$

$$R_{\max} = A + B$$

$$R_{\min} = |A - B|$$

$$4) |(0.2\hat{i} + 0.6\hat{j} + a\hat{k})| = 1$$

$$\sqrt{(0.2)^2 + (0.6)^2 + a^2} = 1$$

$$a = \sqrt{0.6}$$

$$5) (2\vec{A} + 3\vec{B}) \cdot \vec{A} = 0$$

$$2A^2 + 3\vec{A} \cdot \vec{B} = 0$$

$$2(4 + 9 + 1) + 3(2\alpha - 9 - 4) = 0$$

$$28 + 6\alpha - 39 = 0$$

$$\alpha = \frac{11}{6}$$

$$6) \vec{A} \cdot \vec{B} = \text{Scalar}$$

$$\vec{A} \times \vec{B} = \text{vector}$$

$$(\text{Sclar} \neq \text{Vector})$$

$$7)$$

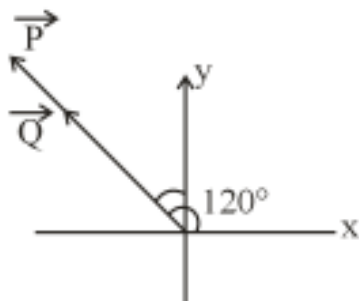
$$\text{Resultant } C = \sqrt{1^2 + 1^2 + 2 \times 1 \times 1 \times \cos(120^\circ)} = 1$$

$$8)$$

Volume is scalar

$$9) \frac{1}{5} = \frac{2}{y} = \frac{1}{z}$$

$$z = 5 \quad y = 10$$



10)

Angle between $\vec{P}$ & $\vec{Q}$ is 0°

$$\square \quad |\vec{R}| = P + Q$$

11)

Explain:

Find the magnitude of the vector $2\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$.

Concept: Pythagorean theorem.

Formula: Magnitude of vector

$$a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = \sqrt{a^2 + b^2 + c^2}$$

$$\text{Calculation : Magnitude} = \sqrt{2^2 + 4^2 + 5^2}$$

$$\text{Magnitude} = \sqrt{4 + 16 + 25}$$

$$\text{Magnitude} = \sqrt{45}$$

$$\text{Magnitude} = 3\sqrt{5}$$

Hence, option (2) is correct.

12)

Angle between $\vec{A}$ and $\vec{B}$

$$180^\circ - 30^\circ = 150^\circ$$

13)

Theory Based

All options are correct.

$$14) R = 2a \cos \frac{\theta}{2}, \quad F = 2F \cos \frac{\theta}{2}, \quad \frac{1}{2} = \cos \frac{\theta}{2}$$

$$\cos 60^\circ = \cos \frac{\theta}{2}$$

$$\therefore \theta = 120^\circ$$

15) (1) Asking About:

Using the triangle law of vector addition, find the correct relationship between the given vectors.

(2) Concept:

The **triangle law of vector addition** states that if two vectors are represented by two sides of a triangle taken in the same order, their resultant is given by the third side in the reverse order.

For the given triangle:

- A. $\vec{C}$ and $\vec{A}$ are in the same order
- B. $\vec{B}$ is in the reverse order.

(3) Formula:

The triangle law of vector addition can be expressed as:

$$\vec{C} + \vec{A} = \vec{B}$$

(4) Solution/Explanation:

- A. In the given triangle, vectors $\vec{C}$ and $\vec{A}$ are aligned in the same cyclic order.
- B. According to the triangle law, the sum of $\vec{C}$ and $\vec{A}$ equals $\vec{B}$ in the reverse direction.
- C. Mathematically:

$$\vec{C} + \vec{A} = \vec{B}$$

- A. Comparing this with the options, **Option 3** is correct.

(5) Final Answer:

Option 3: $\vec{C} + \vec{A} = \vec{B}$

$$16) R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\sqrt{A^2 + B^2} = \sqrt{(A+B)^2 + (A-B)^2 + 2(A+B)(A-B) \cos \theta}$$

$$A^2 + B^2 = A^2 + B^2 + 2AB + A^2 + B^2 - 2AB + 2(A^2 - B^2) \cos \theta$$

$$A^2 + B^2 = 2(B^2 - A^2) \cos \theta$$

$$\theta = \cos^{-1} \left[\frac{A^2 + B^2}{2(B^2 - A^2)} \right]$$

17)

Sum and difference is equal for unit vectors so angle will be 90° .

$$18) \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$(1 - \sin^2 \alpha) + (1 - \sin^2 \beta) + (1 - \sin^2 \gamma) = 1$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

$$19) \cos \alpha = \frac{2}{\sqrt{45}}, \cos \beta = \frac{4}{\sqrt{45}}; \cos \gamma = \frac{-5}{\sqrt{45}}$$

20)

$$ab \sin \theta = 8$$

$$(2)(5) \sin \theta = 8$$

$$\sin \theta = 4/5 \Rightarrow \theta = 53^\circ$$

$$\vec{a} \cdot \vec{b} = ab \cos 53^\circ = 6$$

$$21) \vec{A} \times \vec{B} \perp \vec{C}$$

$$22) \cos \theta = \frac{\vec{P} \cdot \vec{Q}}{P \cdot Q}$$

$$\text{take } \vec{P} = \vec{A} + \vec{B} \text{ and } \vec{Q} = \vec{A} - \vec{B}$$

$$\vec{P} \cdot \vec{Q} = (\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{B}$$

$$\vec{P} \cdot \vec{Q} = 0 \quad \left\{ |\vec{A}| = |\vec{B}| \right\}$$

$$\text{So, } \cos \theta = 0$$

$$\text{and } \theta = 90^\circ$$

23) Vector $(\vec{B} \times \vec{A})$ is perpendicular to the plane containing the vectors $\vec{A}$ and $\vec{B}$. Therefore $\vec{B} \times \vec{A}$ is $\perp \vec{A}$

$$(\vec{B} \times \vec{A}) \cdot \vec{A} = |\vec{B} \times \vec{A}| \cdot |\vec{A}| \cos 90^\circ = 0$$

$$24) \vec{AC} = (2\hat{i} + 4\hat{j} - 5\hat{k}) + (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= 3\hat{i} + 6\hat{j} - 2\hat{k}$$

$$|\vec{AC}| = 7 \text{ unit}$$

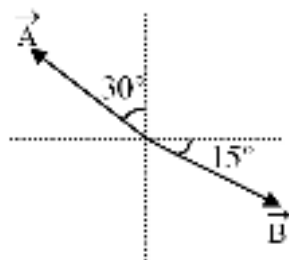
25)

Unit vector perpendicular to $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

$$\frac{\vec{A} \times \vec{B}}{AB \sin \theta} = \hat{n}$$

$$\hat{n} = \frac{\vec{A} \times \vec{B}}{AB \sin \theta}$$



26)

Angle between $\vec{A}$ and $\vec{B}$ is 135° , and resultant will always lie between them. According to option south is not possible.

27) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$

$\vec{F}_3 = -(\vec{F}_1 + \vec{F}_2) = -(4\hat{i} + 6\hat{j})$

28) $\sqrt{4^2 + (4x - 2)^2 + 2^2} = 2\sqrt{1^2 + x^2 + 3^2}$

$\Rightarrow 3x^2 - 4x - 4 = 0$

$\Rightarrow x = -2/3 \text{ or } 2$

29)

As $\vec{A} + \vec{B} = \vec{C}$

$|\vec{A}| = 5$

$|\vec{B}| = 12$

$|\vec{C}| = 13$

$(13)^2 = (5)^2 + (12)^2 + 2(5)(12)\cos\theta$

$\cos\theta = 0$

$\theta = 90^\circ$

30) $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$

$|\vec{A}||\vec{B}|\sin\theta = \sqrt{3}|\vec{A}||\vec{B}|\cos\theta$

$\tan\theta = \sqrt{3} \Rightarrow \theta = 60^\circ$

$|\vec{A} + \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta}$
 $= (A^2 + B^2 + AB)^{1/2}$

31) Given vectors, $\vec{A} = 5\hat{k}$

$\vec{B} = -a\hat{k}$

According to the problem both vectors are equal, then $A = B$, so that the value of $a = -5$.

32) $\vec{V} = \vec{\omega} \times \vec{r}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$$

$$\begin{aligned}
 &= \hat{i} \begin{vmatrix} -2 & 3 \\ 1 & 1 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} \\
 &= \hat{i}(-2-3) - \hat{j}(1-3) + \hat{k}(1+2) \\
 &= -5\hat{i} + 2\hat{j} + 3\hat{k}
 \end{aligned}$$

33) $\vec{A} = 3\hat{i} + 4\hat{j}$

$\vec{B} = \hat{i} + \hat{j}$

Component of $\vec{A}$ along $\vec{B}$ is

$(A \cos \theta)\hat{B}$

$= \frac{\vec{A} \cdot \vec{B}}{B} \hat{B}$

$= \frac{7}{\sqrt{2}} \times \frac{\hat{i} + \hat{j}}{\sqrt{2}} = \frac{7}{2}(\hat{i} + \hat{j})$

34)

A vector can split into maximum 2 rectangular component in its own plane.

35) $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$

Here, the angle θ between the two vectors is

$\theta = 180^\circ - 60^\circ = 120^\circ$

So, $R = \sqrt{(10)^2 + (10)^2 + 2 \times 10 \times 10 \times \cos 120^\circ}$

$$\begin{aligned}
 &= \sqrt{100 + 100 - 100} & \left(\because \cos 120^\circ = -\frac{1}{2} \right) \\
 &= 10 \text{ dyne}
 \end{aligned}$$

36)

Let new vector $\vec{C} = 5|\vec{B}|\hat{A}$

$= 5\sqrt{20} \left(\frac{4\hat{i} + 3\hat{j}}{5} \right)$

$= \sqrt{20} (4\hat{i} + 3\hat{j})$

37)

Resultant makes larger angle with the vector having smaller magnitude.

38) $\vec{A} - 2\vec{B} + 3\vec{C} = (2\hat{i} + \hat{j}) - 2(3\hat{j} - \hat{k}) + 3(6\hat{i} - 2\hat{k})$

$= 20\hat{i} - 5\hat{j} - 4\hat{k}$

$$39) |4\hat{A} + 3\hat{B}| = \sqrt{(4)^2 + (3)^2 + 2(4)(3)\cos 120^\circ}$$

$$= \sqrt{16 + 9 - 12} = \sqrt{13}$$

$$40) \vec{A} = \hat{j} + 3\hat{k}$$

$$\vec{B} = \hat{i} + 2\hat{j} - \hat{k}$$

Area of parallelogram is

$$|\vec{A} \times \vec{B}|$$

$$|\vec{A} \times \vec{B}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 3 \\ 1 & 2 & -1 \end{vmatrix}$$

$$\hat{i}(-1-6) - \hat{j}(0-3) + \hat{k}(0-1)$$

$$-7\hat{i} + 3\hat{j} - \hat{k}$$

$$|\vec{A} \times \vec{B}| = \sqrt{49 + 9 + 1}$$

$$= \sqrt{59} \text{ sq. unit}$$

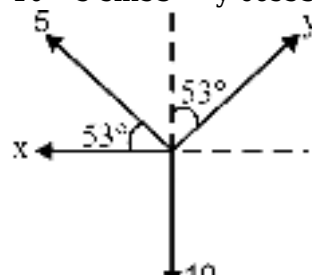
$$41) \Sigma F_x = 0$$

$$\Rightarrow x + 5 \cos 53^\circ - y \sin 53^\circ = 0$$

$$\Rightarrow x + 3 - \frac{4y}{5} = 0 \Rightarrow 5x + 15 - 4y = 0$$

$$\Sigma F_y = 0$$

$$10 - 5 \sin 53^\circ - y \cos 53^\circ = 0$$



$$\Rightarrow 10 - 4 - \frac{3y}{5} = 0 \Rightarrow 6 = \frac{3y}{5} \Rightarrow y = 10$$

$$\square \quad x = 5.$$

$$42) \vec{A} + \vec{B} = 10\hat{i} + 9\hat{j}$$

$$(6\hat{i} + 6\hat{j}) + \vec{B} = 10\hat{i} + 9\hat{j}$$

$$\vec{B} = 4\hat{i} + 3\hat{j}$$

$$|\vec{B}| = 5$$

$$43) |\vec{A} \times \vec{B}|^2 + |\vec{A} \cdot \vec{B}|^2 = A^2 B^2 \sin^2 \theta + A^2 B^2 \cos^2 \theta$$

$$= A^2 B^2$$

$$44)$$

$$\text{As } \vec{R} = \vec{A} + \vec{B} = 5\hat{i} + 5\hat{j} + 5\hat{k}$$

So $\vec{R} = 5\vec{A}$ hence $\vec{R}$ is in the direction of $\vec{A}$

45)

Question Asking About:

The problem involves vectors satisfying the condition $\vec{A} + \vec{B} + \vec{C} = 0$, Given the magnitude of two vector are equal and the third has a magnitude $\sqrt{2}$ time either of them, we need to calculate the angles between the vectors.

The solution relies on:

1. Vector addition for $\vec{A} + \vec{B} + \vec{C} = 0$, $\vec{C} = -(\vec{A} + \vec{B})$.
2. Magnitude Using $|\vec{A}| = |\vec{B}| = x$ and $|\vec{C}| = \sqrt{2}x$
3. Angle relation form the dot product formula

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos \theta \Rightarrow \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}$$

4. The geometry of the problem suggests the vectors form a triangle as their sum is zero.

Solution/Explanation/Calculation:

Step-1 : Write the magnitude condition

from $|\vec{A}| = |\vec{B}| = x$ and $|\vec{C}| = \sqrt{2}x$

$$|\vec{C}| = |\vec{A} + \vec{B}| = \sqrt{\vec{A}^2 + \vec{B}^2 + 2(\vec{A} \cdot \vec{B})}$$

substitute $|\vec{A}|^2 = x^2$ and $|\vec{B}|^2 = x^2$

$$(\sqrt{2}x)^2 = x^2 + x^2 + 2(\vec{A} \cdot \vec{B})$$

Simplify,

$$2x^2 = 2x^2 + 2(\vec{A} \cdot \vec{B}) \Rightarrow \vec{A} \cdot \vec{B} = -\frac{x^2}{2}$$

Step 2: Find the angle between A and B

$$\text{From } \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$$

$$\cos \theta = \frac{-\frac{x^2}{2}}{x \cdot x} = -\frac{1}{2}$$

$$\text{Thus: } \theta = \cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ.$$

Step 3: Find angles involving C

Since $\vec{C} = -(\vec{A} + \vec{B})$, the between C and A or B depends on the symmetry to geometry

Angle between A and C : 135°

Angle between B and C : 135°

Conclusion:

Hence, option (4) is correct

CHEMISTRY

46)

Question Explanation :

200 g of CaCO_3 is heated and decomposes into CaO and CO_2 . We need to find the weight of the residue which is CaO .

Concept :

Use the molar masses and stoichiometry of decomposition.



Solution :

Molar mass of $\text{CaCO}_3 = 40 (\text{Ca}) + 12 (\text{C}) + 3 \times 16 (\text{O})$

$= 100 \text{ g/mol}$

Molar mass of $\text{CaO} = 40 (\text{Ca}) + 16 (\text{O}) = 56 \text{ g/mol}$

From 100 g CaCO_3 , we get 56 g CaO residue.

So, from 200 g CaCO_3 , CaO residue $= \frac{56}{100} \times 200 = 112 \text{ g}$

Final Answer :

Option (4)

47)

Weight of 1 $e^- = 0.00054 \text{ amu}$

Weight of 1 mole $e^- = 0.00054 \text{ g}$
 $= 0.54 \text{ mg}$

$$48) \frac{d_{\text{O}_3}}{d_{\text{O}_2}} = \frac{M_{\text{O}_3}}{M_{\text{O}_2}} = \frac{3 \times 16}{2 \times 16} = 1.5$$

49) Isobars have same mass number but different number of protons.

50)

No. of molecules = Moles $\times N_A$

(1) $\frac{1}{32} N_A$, (2) $\frac{1}{2} N_A$, (3) $\frac{1}{48} N_A$, (4) $\frac{1}{17} N_A$

51)

Nuclear reactions may have loss of mass.

52)

$$(1) d = \frac{\text{MM}}{\text{volume}}$$

$$(2) \text{Molecules} = \frac{18 \text{ mL} \times 1 \text{ g/mL}}{18} \times N_A$$

$$(3) \text{Wt.} = 2 \times 24 = 48 \text{ g}$$

$$(4) \text{No. of H-atoms} = \frac{18}{18} N_A \times 2$$

$$53) M_{wt} = \frac{32 \times 1}{8} \times 100 = 400$$

54)

	O_2	N_2
	$\frac{w}{32}$	$\frac{4w}{28}$
Moles	$= \frac{w}{32}$	$= \frac{4w}{28}$
Molecules	$= \frac{w}{32} N_A$	$= \frac{4w}{28} N_A$
	$= 28 \times 32 \times \frac{w}{32} N_A$	$= 28 \times 32 \times \frac{4w}{28} N_A$
	$= 28 : 32 \times 4$	
	$= 7 : 32$	

55)

$$V = 0.1 \times 22.4$$

$$= 2.24 \text{ L}$$

$$56) n = \frac{11.2}{22.4} = \frac{1}{2}$$

$$M_{wt} = \frac{16}{1/2} = 32$$

$$V.D. = \frac{32}{2} = 16$$

57) Water sample taken from sea, rivers or lake will have molecular formula H_2O . Hence in 18g water, 2g hydrogen is there with 16 g oxygen so the ratio between hydrogen and oxygen is 2 : 16 that is 1 : 8 since the hydrogen and oxygen are in definite weight proportion irrespective of the source from which they obtained.

58)

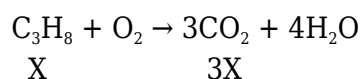
$$\text{No. of atoms} = \frac{16 \text{ amu}}{4 \text{ amu}} = 4$$

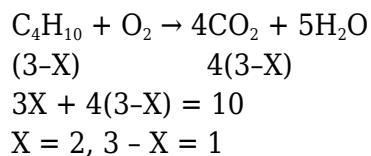
59)

$6e^-$ in C; $8e^-$ in O

$$\text{total number of } e^- \text{ in } CO_2 = 6 + (2 \times 8) = 22$$

60)





61)

$$\text{Wt. of pure CaCO}_3 = 200 \times \frac{90}{100} = 180\text{kg}$$

$$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$$

$$\text{Mole of CaCO}_3 = \text{Mole of CaO}$$

$$\Rightarrow \text{Mole of CaO} = \frac{180 \times 10^3}{100} = 1.8 \times 10^3$$

$$\text{Wt.} = 1.8 \times 10^3 \times 56 = 100.8 \times 10^3 \text{ g}$$

$$= 100.8 \text{ kg}$$

62)

- (1) Wt. of 1 molecule of water = 18 amu
- (2) Number of C atoms = $1 \times N_A \times 6$
- (3) $V = 2 \times 22.4 = 44.8 \text{ L}$
- (4) $V.D. = \frac{M_{wt.}}{2}$

63)

$$\text{No. of atoms} = \text{Molecules} \times \text{atomicity}$$

64)

$$\text{Molecules} = \text{Moles} \times N_A$$

65)

$$\text{Atoms} = \text{Mole} \times N_A \times \text{atomicity}$$

$$1.25 \times 10^{24} = \frac{w}{150} N_A \times 5$$

$$\frac{1.25 \times 10^{24}}{6 \times 10^{23}} \times 30 = w$$

$$w = 62.5 \text{ g}$$

$$66) \quad n = 4, \ell = 2, s = +\frac{1}{2}$$

67) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

68)

Spin Q.N. Can not be calculated.

69)

$$4s \rightarrow n = 4, \ell = 0$$

$$\text{Angular momentum} = n \frac{h}{2\pi} = 4 \times \frac{h}{2\pi} = \frac{2h}{\pi}$$

$$\begin{aligned} \text{orbital angular momentum} &= \sqrt{\ell(\ell+1)} \frac{h}{2\pi} \\ &= 0 \text{ for } \ell = 0 \end{aligned}$$

70)

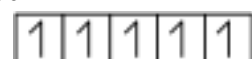
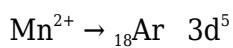
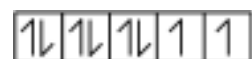
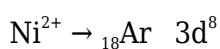
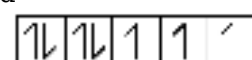
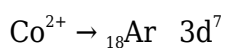
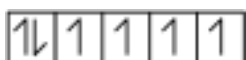
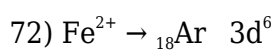
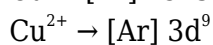
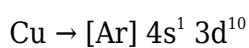
n = energy and size of orbit

ℓ = shape of orbital

m = orientation of e^- cloud

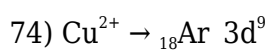
s = spin of e^-

71)



$$73) \Psi_{311} = \Psi_{n,\ell,m}$$

$$n = 3, \ell = 1, m = 1$$



75)

$$\text{Orbital angular momentum} = \frac{h}{2\pi} \sqrt{\ell(\ell+1)}$$

76) For 3p

$$n-1 = 2$$

$$\text{R.N.} = n - \square - 1 = 3 - 1 - 1 = 1$$

$$\text{A.N.} = \square = 1$$

77)

$3p_y$ has xz nodal plane.

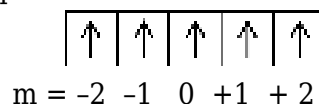
$$78) 0.529 \times \frac{(1)^2}{1} \text{ \AA} = X$$

$$2\pi r = n\lambda$$

$$2\pi \times 0.529 \times \frac{3^2}{1} = 3\lambda$$

$$\boxed{6\pi X = \lambda}$$

79) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$



80) No. of orbitals in an orbit = n^2

$$81) \boxed{2\pi r = n\lambda}$$

'n' is orbit no.

82) $1s^2 2s^2 2p^3$

$$s \text{ } e^- = 4$$

$$p \text{ } e^- = 3$$

83)

For H-atom energy of e^- is same in same subshell of an orbit.

84)

Uncertainty principle :- Exact position and momentum of e^- can't be determined.

$$85) \lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.63 \times 10^{-34} \text{ J. sec}}{10^{-3} \text{ kg} \times 100 \text{ m sec}^{-1}}$$

$$\lambda = 6.63 \times 10^{-33} \text{ m}$$

86)
$$\lambda = \frac{h}{\sqrt{2mKE}}$$

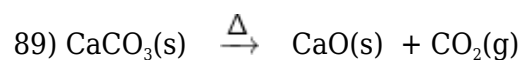
87) No. of e^- in any orbital equal to two.



$$\frac{\text{mole}}{\text{s.c.}} = \frac{5}{1} \quad \frac{8}{2}$$

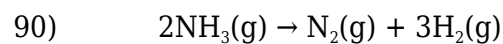
□ B is L.R.

$$\text{mole of C} = \frac{8}{2} = 4 \text{ mole}$$



$$\frac{10^3 \text{ mole}}{\text{wt. of CaCO}_3} = \frac{44 \text{ kg. wt}}{44 \times 10^3} \text{ moles}$$

$$\text{wt. of CaCO}_3 = 10^3 \times 100 \text{ g} = 100 \text{ kg}$$



$$\frac{\text{wt}}{\text{mole}} = \frac{34\text{g}}{17} = 2 \text{ mole}$$

$$\frac{\text{moles of NH}_3}{\text{s.c.}} = \frac{\text{moles of N}_2}{1} = \frac{\text{moles of H}_2}{3}$$

$$\text{Total moles} = 1 + 3 = 4$$

$$\text{Vol at STP} = 4 \times 22.4 = 89.6 \text{ L}$$

BIOLOGY

91)

NCERT, Pg. # 41 to 44

92)

NCERT, Pg. # 43

93)

94) NCERT Pg. No. 44

95) NCERT Page no. 38 & 44

96) The assertion is false because it incorrectly states the levels of organisation in sponges and coelenterates. Sponges have a cellular level of organisation, not a tissue level, as they are composed of loose cell aggregates. Coelenterates, on the other hand, exhibit a tissue level of organisation, as cells performing the same function are arranged into tissues. The reason is true as it correctly describes the organisation of cells in sponges and coelenterates. Therefore, A is false but R is true.

97) The assertion is false because coelenterates do not exhibit bilateral symmetry; they exhibit radial symmetry where the body can be divided into identical parts by any plane passing through the central axis. The reason is true as coelenterates are indeed diploblastic animals with two embryonic layers and mesoglea in between. Therefore, A is false, but R is true.

98)

NCERT-XI, Pg. # 44

99)

Explaining Question:

Amos vertebrates, which of the following do not Exclusively Respire through lungs?

Concept: This question is based on Respiratory organs in organism

Solution: Amphibians do not Exclusively Respire through lungs. Some like frogs can respire through their moist skin also.

100) NCERT-XI, Pg. # 41

101) Explanation

The given organisms belong to different orders of Class Mammalia:

1. Ornithorhynchus (*Platypus*) → Order Monotremata

A. Egg-laying mammal (Prototheria).

2. Macropus (*Kangaroo*) → Order Marsupialia

A. Marsupial (Metatheria).

3. Pteropus (*Flying Fox/Bat*) → Order Chiroptera

A. Placental mammal (Eutheria).

4. Balaenoptera (*Blue Whale*) → Order Cetacea

- A. Largest marine mammal on Earth.
B. Completely aquatic with a streamlined body.

Correct Answer:

(2) Ornithorhynchus, Macropus, Pteropus & Balaenoptera

102)

NCERT, Pg. # 44

103)

NCERT, Pg. # 48

104)

NCERT XI, Pg.#44

105) NCERT (XI) Pg# 40

106) **Explanation:**

Balanoglossus (a marine acorn worm) belongs to the phylum Hemichordata, which shares characteristics with both chordates and non-chordates.

- A. Balanoglossus is a marine animal, not freshwater.
B. Balanoglossus has a buccal diverticulum, which was once mistaken for a notochord but is not a true notochord.
C. Excretion occurs through a proboscis gland, which functions similarly to a kidney.
D. Balanoglossus is triploblastic, but its circulatory system is open, not closed.

Correct Answer:

Option 3. Excretion by proboscis gland

107)

NCERT (XI) Pg. # 52

108)

NCERT (XI) Pg. # 55

109)

NCERT (XI) Pg. # 48

110)

NCERT, Pg. # 38, 39

111)

NCERT Reference: Class XI, Page No. 48,52

112) NCERT, Pg. # 39

113)

NCERT, Pg. # 47, 48

114) NCERT (XI) Page No. # 51

115) NCERT (XI) Page No. # 59

116) NCERT (XI) Page No. # 54

117)

NCERT, Pg. # 41

118) NCERT Pg.#52

119) NCERT Pg.#48

120) NCERT Pg.#51

121) NCERT Pg.#49

122) NCERT Pg.#47

123) NCERT Pg.#51

124) NCERT (XI) Pg # 53

125) NCERT (XI) Pg # 53

126) NCERT (XI) Pg # 54

127) NCERT (XI) Pg # 55

128) NCERT (XI) Pg # 49

129)

NCERT, Pg. # 41

130) NCERT-XI, Page#49

131)

NCERT, Pg. # 80

132)

NCERT, Pg. # 83

133)

NCERT, Pg. # 83

134)

NCERT, Pg. # 84

135) NCERT Pg. # 81 (E)

136) NCERT Pg. # 88

137) NCERT Pg. # 100

138) NCERT, Pg. # 99

139) NCERT, Pg. # 97

140) NCERT (XI) Pg. # 101

141)

Ncert Page 99.

142)

Module No. 1, Pg. # 171

143) **Explaining the Question** - Question asking about the function of plasma membranes.

Concept - This Question is based on Cell membrane **Solution** - As the polar molecules cannot pass through the nonpolar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membrane. A few ions or molecules are transported across the membrane against their concentration gradient, i.e., from lower to the higher concentration. Such a transport is an energy dependent process, in which ATP is utilised and is called active transport, e.g., Na^+/K^+ Pump.

Final Answer - 3

144) NCERT Page 95.

Hints : Ribosomes, Mitochondria and Chloroplast

145) NCERT-XI, Pg. # 99 (Figure - 8.10)

146)

NCERT XI, Pg. # 101

147) NCERT Page No. 97, 98

148)

NCERT-XI, Pg # 95

149)

NCERT XI Pg # 93, 94

150) NCERT Page No. 98.

151) NCERT (XI) Pg. # 101

152) NCERT-XI, Pg. # 93

153) **Explanation**

• In anaphase of mitosis, the sister chromatids separate, effectively doubling the chromosome number temporarily. • Since the G_2 phase has 8 chromosomes and 10 Pg DNA, when the chromatids separate, the cell temporarily has 16 chromosomes but still 10 Pg DNA (since DNA amount does not change, only distribution changes).

Correct answer: 3. Anaphasic cell - 16 chromosomes with 10 Pg DNA.

154) NCERT XI, Page # 124

155)

NCERT-XII Page # 123, 124

156) NCERT XI Pg. # 126

157)

NCERT Pg. # 123

158)

NCERT Pg. # 126

159)

SOLUTION—

Onion root tip cell ($2n = 16$):

- G₁-phase: 16 chromosomes (each with one chromatid)
- Metaphase: 16 chromosomes (each with two sister chromatids)
- Anaphase: 32 chromosomes (as sister chromatids separate and become individual chromosomes)
- Just after M-phase: 16 chromosomes in each of the two daughter cells.

The correct answer is: OPTION(1) 16, 16, 32, 16

160)

NCERT XI pg no.126

161)

NCERT-XI, Pg. # 123, 124

162)

XI NEW NCERT Pg # 123 - 124

163)

SOLUTION-

This sequence represents the order in which a cell grows and divides:

- G₁ phase (Gap 1):
The cell grows and prepares for DNA replication.
- S phase (Synthesis):
DNA replication occurs.
- G₂ phase (Gap 2):
The cell continues to grow and prepares for mitosis.
- M phase (Mitosis):
The cell divides its nucleus (mitosis) and cytoplasm (cytokinesis).

The correct answer is:

OPTION (3) G₁ - S - G₂ - M

164) NCERT-XI, Pg. # 124

165) NCERT, Page - 125, 128

166) NCERT Pg. # 127

167) NCERT XI Pg. No. # 121, 123

168) NCERT Pg. No. : 123

Explanation :

Read statement I & II In question and select the correct option. **Concept:**

Endomembrane system.

Solution/Explanation:

Endomembrane system is formed by membrane-bound organelles, i.e., the endoplasmic reticulum, Golgi complex, lysosomes, and vacuoles, because their functions are coordinated. Mitochondria and chloroplasts are not part of the endomembrane system as they are not coordinated with the above components.

Final Answer :

Option (2)

169)

NCERT Pg. # 126

Explain Question : You're asked to determine which statements about bivalents are correct.

Concept : This Questions based on Prophase I of meiosis

Explanation :

(i) "Bivalents are tetrads" - Correct

A bivalent contains 4 chromatids, so it's also called a tetrad.

ii) "A bivalent means 4 chromatids and 2 centromeres" - Correct

Each of the 2 homologous chromosomes has 2 chromatids and 1 centromere → total: 4 chromatids, 2 centromeres.

(iii) "One bivalent consists of 2 homologous chromosomes" - Correct

That is the definition of a bivalent - The pairing of two homologous chromosomes.

Final answer : 1

170) NCERT Pg. # 125

SOLUTION- Meiosis, a type of cell division, reduces the chromosome number by half to produce haploid gametes (sperm and egg cells). During fertilization, these gametes fuse to form a diploid zygote, restoring the original chromosome number of the species. This process ensures that the chromosome number remains constant across generations in sexually reproducing organisms.

The correct answer is: OPTION(1) Assertion and Reason are true and Reason explains assertion correctly

171) NCERT XI Pg # 123

172) NCERT-XI, Pg # 135, 8.5.4

173) NCERT XI Pg # 138

174) NCERT (XIth) Pg. # 126

175) Centriole

NCERT-XI Pg#126,136,138

176)

NCERT Pg. No. : 95

1. Asking About: Organelle, structure and function.

2. Concept: Golgi complex.

3. Solution/Explanation: Golgi complex is single membrane bound in which many flat, disc shaped cisternae modify chemicals and synthesize glycolipids and glycoproteins.

177)

NCERT Pg. # 125

178) NCERT Pg. # 99

179) NCERT, Pg. # 123

180) NCERT, Pg. # 97